

The two wheels of a bullock cart



The Olympic rings

Fig. 4.50

Thinking Corner

Draw four congruent circles as shown. What do you infer?



Position of a Point with respect to a Circle

Consider a circle in a plane (Fig.4.51). Consider any point P on the circle. If the distance from the centre O to the point P is OP, then

(i) OP = radius (If the point P lies on the circle)

(ii) OP < radius (If the point P Point lies inside the circle)

(iii) OP > radius(If the point P lies outside the circle)

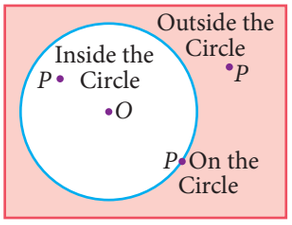


Fig. 4.51

So, a circle divides the plane on which it lies into three parts.

Progress Check

Say True or False

1. Every chord of a circle contains exactly two points of the circle.
2. All radii of a circle are of same length.
3. Every radius of a circle is a chord.
4. Every chord of a circle is a diameter.
5. Every diameter of a circle is a chord.
6. There can be any number of diameters for a circle.
7. Two diameters cannot have the same end-point.
8. A circle divides the plane into three disjoint parts.
9. A circle can be partitioned into a major arc and a minor arc.
10. The distance from the centre of a circle to the circumference is that of a diameter

Thinking Corner

1. How many sides does a circle have ?
2. Is circle, a polygon?

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